

PAGE 2026 — {ospsuite} + {esqlabsR} Cheatsheet

Workshop reference — 2 pages

Pavel Balazki · Wilbert de Witte

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{ospsuite} essentials

Workflow

load → inspect → modify → run → extract → analyze
→ plot

Load + save

```
sim <- loadSimulation("path/Model.pkml")
saveSimulation(sim, "path/Model_v2.pkml")
# Independent copy? Reload
sim_copy <- loadSimulation("path/Model.pkml")
```

Inspect

```
getAllParameterPathsIn(sim)
getAllMoleculePathsIn(sim)
getAllContainerPathsIn(sim)
getAllParametersMatching("**|Volume", sim)
```

Parameter access

```
p <- getParameter("Organism|Liver|Volume", sim)
p$value; p$unit
setParameterValuesByPath("Organism|Weight", 75, sim, units =
  ↪ "kg")
```

Run

```
res <- runSimulations(sim)[[1]] # single
res_list <- runSimulations(list(sim1, sim2)) # multiple
```

Extract

```
out <- getOutputValues(res, quantitiesOrPaths =
  ↪ "...|Concentration")
out$data # → data.frame: Time, paths, values
```

Plot — DataCombined

```
dc <- DataCombined$new()
dc$addSimulationResults(res)
dc$addObservedData(loadDataSetFromPKML("obs.pkml"))
plotTimeProfile(dc) # time profile
plotPredictedVsObserved(dc) # GoF
plotResidualsVsCovariate(dc, xAxis = "time") #
  ↪ residuals
```

PK params

```
pk <- calculatePKAnalyses(res, "...|Concentration")
pkAnalysesAsDataFrame(pk) # AUC, C_max, t_max, t½
addUserDefinedPKParameter(...)
```

Local sensitivity

```
sa <- sensitivityCalculation(
  simulation = sim,
  variableParameterPaths = c("...|Lipophilicity",
    "...|GFR_filtration_fraction"),
  variationRange = 0.1)
mostSensitiveParameters(sa, totalSensitivityThreshold = 0.9)
```

Units

```
toUnit("Concentration", 1, "mg/L")
toBaseUnit("Volume", 70, "kg")
toDisplayUnit(p, p$value)
```

Path conventions

Aciclovir|Organism|Liver|Volume

Levels: organ → compartment → parameter. | separates levels. ** = recursive wildcard.

Parameter identification

```
library(ospsuite.parameteridentification)
```

Optimization parameter

```
piPar <- PIParameters$new(
  parameters = getParameter("Aciclovir|Lipophilicity", sim))
piPar$minValue <- -5
piPar$maxValue <- 5
piPar$startValue <- 2 # optional
```

Observed data → output mapping

```
obs <- loadDataSetsFromExcel(
  "Data/Obs.xlsx", projectConfiguration = project)

simPath <- "Organism|PeripheralVenousBlood|Aciclovir|Plasma
  ↪ (Peripheral Venous Blood)"
mapping <- PIOutputMapping$new(
  quantity = getQuantity(simPath, container = sim))
mapping$addObservedDataSets(obs$`Aciclovir.Vergin1995.IV250`)
```

Configure + run

```
cfg <- PIConfiguration$new()
cfg$algorithm <- "DEoptim" # or "LevenbergMarquardt"
cfg$autoEstimateCI <- TRUE
```

```
piTask <- ParameterIdentification$new(
  simulations = sim,
  parameters = piPar,
  outputMappings = mapping,
  configuration = cfg)
```

```
res <- piTask$run()
piTask$plotResults()
```

{esqlabsR} essentials

Project structure

```
MyProject/
  ProjectConfiguration.xlsx      ← master
  Models/
    Simulations/      *.pkml
  Configurations/
    Scenarios.xlsx
    ModelParameters.xlsx
    Applications.xlsx
    Plots.xlsx
  Data/                *.xlsx
  Results/
  Reports/              *.qmd
```

Initialize

```
library(esqlabsR)
initProject(path = "~/MyProject")
```

Project object

```
project <-
  ↳ createProjectConfiguration("ProjectConfiguration.xlsx")
project$paths$ModelsFolder
project$paths$ConfigurationsFolder
```

Load + run scenarios

```
cfg <- readScenarioConfigurationFromExcel(
  scenarioNames = c("PO_7.5mg", "IV_2mg"),
  projectConfiguration = project)
sc <- createScenarios(cfg)
res <- runScenarios(scenarios = sc)
```

Snapshot / restore

```
snapshot(project)      # Excel → JSON snapshot
restore(project)        # JSON → Excel
status(project)         # diff
```

Plots from Excel

```
plots <- createPlotsFromExcel(
  plotGridNames = c("PO_vs_obs"),
  simulatedScenarios = res,
  observedData = obs,
  projectConfiguration = project)
plots[[1]]
```

Sensitivity (multi-range)

```
simulation <- scenarios$PO_7.5mg$simulation
analysis <- sensitivityCalculation(
  simulation = simulation,
  outputPaths = c(Plasma = "..."),
  parameterPaths = c(Lipo = "Midazolam|Lipophilicity"),
  variationRange = c(0.1, 0.5, 1, 2, 10),
  pkParameters = c("C_max", "AUC_inf"))
sensitivitySpiderPlot(analysis)
sensitivityTornadoPlot(analysis)
sensitivityTimeProfiles(analysis)
saveSensitivityCalculation(analysis, outputDir = "Results/SA")
```

Excel cheat-row reference

File	One row =	Key columns
Scenarios	one scenario	Scenario, ModelFile, ApplicationProtocol, IndividualId, OutputPathsId
ModelParameters	parameter override	Id, ContainerPath, ParameterName, Value, Units
Applications	dosing protocol	Id, DrugMass, DoseUnit, StartTime, Route
Plots	one figure	PlotID, ScenarioName, OutputPath, xLog, yLog

Quarto report — re-render

```
quarto render Reports/MyReport.qmd      # default
quarto render Reports/MyReport.qmd -P dose_mg:15  # override
  ↳ param
quarto render Reports/MyReport.qmd --to pdf      # PDF only
```

In a chunk

```
# parameters from YAML
params$dose_mg

# auto-run scenarios
cfg <- readScenarioConfigurationFromExcel(params$scenario,
  ↳ project)
sc <- createScenarios(cfg)
res <- runScenarios(sc)
plots <- createPlotsFromExcel(plotGridNames = "MyPlot",
  simulatedScenarios = res,
  projectConfiguration = project)
```

Repo

r-training.esqlabs.com github.com/esqLABS/r-
packages-training